

Exploring the Difference of Attention, Emotion, and Neurobiological Domains as Underlying Vulnerabilities For Game Addiction

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ABSTRACT

Recently, game addiction has officially been considered as a mental illness by WHO because of its negative consequences on people's health. However, the predictive factors that make some individuals more at risk for game addiction are not well recognised. The aim of the present study was to investigate whether levels of attention, emotional responses, perceived immersion, and specific brain activities might be associated with the development of game addiction. A total of thirty-six participants were separated into two different groups based on high or low vulnerability to game addiction. During the experiment, frontal EEG signals were acquired while the participants were playing VR game and while other relevant behavioural tasks were completed. Results showed that the high vulnerability group had faster reaction time in pre- and post- gaming flanker tasks and a lower absolute beta power in comparison to the low vulnerability group. The current study accentuates the significance for future investigations on constructing complete pictures of game addiction and provides new insights into the diagnosis and treatment of game addiction. Limitations and future implications of the study were also discussed.

KEYWORDS

Game addiction, Virtual Reality, Electroencephalogram, Emotion, Selective Visual Attention, Digital Health

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1 INTRODUCTION

Gaming has become a popular leisure activity in many parts of the world among adolescents and young adults. According to the UK Interactive Entertainment Association, in 2018, there were approximately 37.3 million people in the UK that play games,[11] which was more than half of the country's total population. While there are various positive effects of gaming with regard to clinical and

educational fields [65], an increasing number of empirical studies examine the small minority that appears to develop problems as a result of excessive gaming.[9] Indeed, game addiction has been suggested to be a mental health problem. Evidence suggests problematic gameplay may further impair everyday life, academic attainment, social relationships, and emotional development, particularly among adolescents.[9] Although many studies have attempted to explore the cause of game addiction in terms of demographic factors, neurological conditions, psychosocial, and cognitive functions, [39,52] the causes or aggravating factors that lead to game addiction still remain uncertain. Moreover, there is comparatively little HCI research that compares the underlying vulnerabilities and the prevalence of individuals who display addictive gaming tendencies with other individuals who do not display addictive behaviours.

Therefore, it might be beneficial to compare the differences in behaviours and addictive potentials between game addicted and non-game addicted individuals. Consequently, new insights could be added into this field that may help in offering guidance on individuals who are at risk of becoming addicted gamers so they can be identified and treated at an earlier stage. The current study aims to investigate whether there are differences in attention, emotional responses and perceived immersion between high vs. low vulnerability game addicts groups during gameplay, by adopting behavioural and neurobiological measures.

2 LITERATURE REVIEW

2.1 Background of Game Addiction

With the continuously advancing development of technology in the 1990s, video had come to play a crucial part of leisure culture in a way that became a matter of concern for some researchers[16] Apart from solely being used for entertainment purposes, gaming has also been found to be effective as an educational or therapeutic tool that could relieve everyday stress, enhance cognitive and motor skills, and even bring out positive behavioural in psychotherapy patients.[17,22,40] On the other hand, along with the increasing popularity and impact around the world, gaming has often negatively linked with problematic, excessive and addictive behaviours with adverse consequences. [6,10] Although it only appears to affect a minority of game users, the addictive game usage has caused many to have health issues, social impairments, and decreased academic attainment [31,65].

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also been found to be effective as an educational or therapeutic tool that could relieve everyday stress, enhance cognitive and motor skills, and even bring out positive behavioural in psychotherapy patients.[17,22,40] On the other hand, along with the increasing popularity and impact around the world, gaming has often negatively linked with problematic, excessive and addictive behaviours with adverse consequences. [6,10]Although it only appears to affect a minority of game users, the addictive game usage has caused many to have health issues, social impairments, and decreased academic attainment [31,65].

In 1989, the first empirical study on game addiction was published by Shotton[58], in which she recruited a relatively small sample of 127 people that mainly comprised young adult males with self- described “hooked” behaviours on video games for a minimum of 5 years. Shotton later conceptualised game addiction to be more positive than negative, and reported that participants in her study seem to be more intelligent and motivated but were misunderstood in a negative way by others in society. However, although this study provided first insights and pioneered in the investigation of the potential impact of game addiction, one main drawback was that no standardised measure of game addiction was used. The only criteria used in this study to identify the “game addicted” was merely self- reported gaming experiences without further validation or research. Therefore the results might not be seen as an accurate representation of game addicted group. Despite this, a number of researches pointed out that an individual’s self-diagnosis of their game addiction condition was often found to be correlated with more standardised measures of game addiction. [69]Since then, game addiction has received increasing public attention and started to be a highly debated topic among health authorities, gamers, and researchers.

Therefore, although it is evident that gaming addiction can occur and have negative consequences further damaging individuals’ daily lives, the uncertainty still remains on whether game addiction should be considered as a behavioural addiction or mental disorder. Further research is needed in this area to help with the understanding of game addicts population and its nature and severity.

2.2 Negative Consequences of Game Addiction

Notwithstanding of whether problematic gaming can be categorised as an addiction, there has been substantial growth in the number of studies that indicate excessive gaming can result in a wide-ranging variety of deleterious psychosocial consequences for a minority of addicted individuals. These comprise poor performances in work [70], academic studies [12], socialising, time with family, and sleep [48]. Moreover, addictive gaming behaviours might lead to decreased life satisfaction [35], worse psychosocial well-being[27], attention issues [25], increased aggressive behaviours [19], decreases in verbal memory performance [10], and increased depression and anxiety [52]. It is important to highlight that time spent playing video games alone was not linked with those negative social, emotional, and mental outcomes and that these negative outcomes are specifically related to video game addiction [3]. In addition, many research highlighted some health and medical consequences that may cause by problematic gaming. These include epileptic seizures [57], hearing delusions [60], obesity [56], pain

in the wrist or neck, and numb fingers. [42,43,63] In conclusion, the possible adverse consequences for psychosocial and mental health mentioned in the studies clearly emphasised the importance and urgency to conduct more research in this field to build more extensive acknowledgement for the medical community to prevent or treat game addicts

2.3 The use of neuroimaging techniques in game addiction studies

In the past decade, numerous empirical studies were used to shed light upon the issue of game addiction from a neuroscientific perspective by adopting neuroimaging techniques [30]. This is because only with neuroimaging approaches, it became possible to distinguish specific brain areas that are related to the development of game addiction. Additionally, the neurobiological latent factors behind these mechanisms and environmental risk factors also play important roles in explaining the reason why some individuals are more vulnerable to game addiction than others. For example, through the use of Functional Magnetic Resonance Imaging (fMRI), it has been illustrated that game addicts tend to display similar neural processes and increased activity in brain areas associated with substance-related addictions and other behavioural addictions, such as obsessive gambling [28]. Meanwhile, it has also been reported that those with game addiction tends to have a higher prevalence of two specific polymorphisms of the dopaminergic system which suggests that among some players there might be some genetic predisposition to develop video game addiction [23]. Moreover, by using electroencephalogram (EEG) on excessive gamers in an attempt to learn about their error- processing and response inhibition, Littel et al [38] found that excessive gamers had reduced frontal-central Event Related Potential amplitudes following incorrect responses, in comparison to correct responses. This has led to poor error- processing in comparison to controls. Moreover, they also reported that excessive gamers also exhibited less inhibition on both self-report and behavioural measures.

Numerous studies were conducted to investigate the relationship between emotion and the brain. To name a few, Bos [1] indicated that left frontal inactivation was linked with negative emotion whereas right frontal inactivation was related to positive emotion. In supportive of this, joy and happiness were found to be associated with comparatively greater left frontal alpha activation while fear and sadness demonstrated greater activation in right frontal alpha [66]. To conclude, reflection of emotions could be observed from the asymmetry of inactivity in left or right brain by using alpha or beta wave as measurement.

2.4 Absolute Beta Power as Indicator for Game Addiction

Choi et al [59] carried out a neuroimaging study on Internet Gaming Disorder (IGD) patients and their findings revealed a lower absolute beta power in the resting state of individuals with IGD. Later these results were replicated by Lee et al [34] with another comparison group: IGD individuals with depression and without depression. Based on their analysis, the IGD group showed lower absolute beta power when compared with the IGD-Depression group, implying that absolute beta power could be a potential biomarker of internet

game addiction. In the present study, we also use beta waves as a neurobiological measurement for game addiction vulnerability. So that brain activity comparison could be shown between High Vulnerability Game Addicts (HVGA) and Low Vulnerability Game Addicts (LVGA). Doing a between-group comparison would enable the observation of the varied extent of brain activity between groups and help to understand why neurocognitive domain are more linked with game addiction.

2.5 Aims and Hypothesis of Current Project

The review of literature has highlighted potential links between game addiction and the associated risk factors like attention, emotional responses and perceived immersion. The enhanced cognitive ability after gaming was also discussed. However, the distinctive features that separate game addicts from healthy controls are not well recognised. Therefore, the aim of this study was to identify underlying factors that may have an impact on the development of gaming addiction.

Based on the research analysed, the following hypotheses are formulated to illustrate the difference in task performance between individuals with higher vulnerability to game addiction and those with lower vulnerability.

1. Differences in absolute beta power between HVGA and LVGA, with lower absolute beta power in HVGA group.
2. Differences in emotional responses between HVGA and LVGA, with greater emotional response in HVGA group.
3. Differences in perceived immersion between HVGA and LVGA, with higher perceived immersion in HVGA group.
4. Better selective visual attention performances in pre- and post-gaming attention test, with HVGA group reacted faster than LVGA group in both conditions.
5. Difference in number of participants requesting for extra play-time, with more requests from HVGA group.

3 METHODOLOGY

3.1 Design

The study adopted a between-subjects design (Refer to Figure 1). Between-subjects comprise two different groups of HVGA and LVGA, who will complete pre- and post- gaming session attention tasks, with pupil size and an EEG measured during gameplay. All trials in attention tasks and items in questionnaires were randomised. Counterbalancing for the order of attention tasks was applied. Different stimuli were used in pre- and post- gaming sessions to the effect of practice and familiarity.

3.2 Participants

Twenty-six participants (15 females and 11 males) ranging from 18 – 25 years took part in an approximate 15-minute Virtual Reality laboratory session and a short post-experiment experience survey. All of the participants were university students and residents in London at the time of the study. Participants were recruited through volunteer sampling techniques using the UCL Psychology Subject Pool (SONA), and through posts on 3 social media platforms: Facebook, Instagram, and WeChat. To be eligible for the laboratory session, participants had to fill out an online questionnaire consisting of questions regarding Internet Gaming Disorder Scale-Short

Form 9 and their previous gaming experiences. The results were then used for pre- screening the participants, where those with scores from 20 to 24 were excluded for subsequent studies. Thirteen participants with pre- screening scores of 25 and above were assigned to the High Vulnerability Game Addicts (HVGA) group and 13 participants with scores of 19 or below were assigned to the Low Vulnerability Game Addicts (LVGA) group. Each participant was offered a £10 Amazon voucher or 0.5 SONA credit at the completion of the experiment as an incentive. Ethical approval for the study was granted by UCL Department of Information Study Ethics Chair (Project ID No. DIS21112018) with the UCL Data Protection Registration Number (Z6364106/2018/11/52).

3.3 Materials

Online questionnaires and surveys Online questionnaire tool was used to construct and deliver the pre-screening and post-experiment questionnaires. The full survey can be found in Appendix. Pre-screening questionnaire: this questionnaire begins with self-reports of previous gaming experiences and background questions including factors such as age, gender, frequency of gaming, length of each play session, and estimated gaming time (hours/week). Then followed by the IGDS-SF9 for assessing IGD severity and effects. Post-experiment questionnaire: this questionnaire was constructed to assess the level of immersion and mood reaction while playing the VR game.

Cognitive ability task The selective visual attention test program (Eriksen Flanker task) was designed and assessed online via computers using Gorilla software (V1.0.gorilla Sc). Accurate response rate and reaction times were recorded by the software.

EEG recording EEG data was recorded by LooxidVR headset, it provides 9 flexible PCB-based EEG electrodes (AF3, AF4, AF7, AF8, Fp1, Fp2, 2 GND, 1 Ref). It acquires 6-channel, 24-bit EEG data with sampling rates up to 1KHz. Different sizes of the EEG cap were used according to the subject's forehead size. Please see appendix for more product details.

Virtual Reality game The game Virtual Virtual Reality (1.7.16) by Tender Claws was selected to be played by participants during recording sessions. It is a single-user narrative-drive adventure game about VR and AI and was rated the highest on Google Daydream platform.

3.4 Measures

Self-report measures

Demographics and previous gaming experiences The demographics and previous gaming experiences survey measured participants' gender, age group, and duration of gaming behaviours.

Internet Gaming Disorders Scale -Short Form (IGDS-SF9) A nine-statement questionnaire with a 5-point Likert scale (Never, Rarely, Sometimes, Often, Very Often) was used for measuring IGD severity. [49] Responses were converted into scores to distinguish between HVGA and LVGA. Minimal adaptations were made to wording to enhance clarity and for use of the questionnaire for gaming topic.

Immersive Experience Questionnaire The immersive experience questionnaire used was adopted from the "Immersion in Game

Scale” in Cox et al [8], and takes approximately 4 minutes to complete. It consisted of 17 items related to experiencing VR game play with a 5- point Likert scale (1= not at all and 5=very much so) to indicate how the participants felt at the end of the game. (Refer to Appendix for more details) The questionnaire was also modified to refer to the VR game particularly. Irrelevant items were removed. Minimal adaptations were made to wording with the aim to reduce complexity for non- gamers. Immersion scores are calculated by adding up participants’ answers to all questions. Scherer et al [53] and takes about 3 minutes to complete. (Refer to Appendix for more details) Participants were asked to indicate the emotion they were experiencing by choosing intensities (0=None and 5=very much so) for a single emotion or a blend of several emotions out of 20 distinct emotion families. The format of the survey was adapted from wheel-like to a list of emotions with sliders on the side to improve clarity and reduce operation difficulty on computers.

Behavioural measures

Eriksen Flanker task The Eriksen Flanker task was selected as the cognitive task for measuring selective visual attention performance before and after gaming. During the task, participants were instructed to only focus on the target stimuli in the middle while ignoring adjacent flanker stimuli. All stimulus may be facing the same direction (congruent condition), or the centre stimuli may be facing the opposite direction (incongruent condition). Instructions were to press the corresponding direction key (left arrow key or right arrow key) depending on the direction of the centre stimuli while responding as fast as possible. Before each trial, a fixation cross was displayed in the centre for a random order of either 400ms, 800ms, or 1200ms. Correct/incorrect response feedback was provided to participants after completion of each trial. Reaction time and response accuracy were collected. The pre-gaming flanker task comprised 4 practice trials and 12 test trials (6 congruent trials and 6 incongruent trials) with fish as the stimuli, where the post-gaming flanker task comprised 20 test trials (10 congruent trials and 10 incongruent trials) with arrow as the stimuli.

EEG measures EEG recordings files were first converted from lx. format to csv. Format. Then all files were transferred to Brainstorm MATLAB, with a sampling frequency rate at 1000 Hz. Inspection of bad channel were made by down-sampling all data to 250Hz. Participants with poor EEG data quality were excluded for subsequent analysis. Band-pass filters were used for pre-processing total (0.5-35 Hz) and beta waves (15-29 Hz) correspondingly. Time- Frequency analysis was carried out by using Fourier Transformation. New data files were generated to display absolute power values for beta and total waves for comparison between groups.

3.5 Procedure

The study comprised of four parts: firstly, participants provided informed consent and completed the online pre-screening survey, and those who met the prerequisites were allocated to HVGA or LVGA group and were then invited to the laboratory. For the second part, upon arrival, an information sheet and consent form were provided for the participants to read and sign, which offered information about the study and the session, and informed and assured participants that their data was anonymised. Before the beginning of the session, provided each participant signed the consent form

and agreed to proceed, participants were informed on the topics the experiment was formulated around”, and questions raised at this time were answered to ensure the participant was fully informed and comfortable in continuing with the study. At this point the participants were asked to complete the pre-gaming visual selective attention task using a laptop computer. In the third part, LooxidVR cap was fitted onto each participant in the experiment room, EEG signals and eye-tracking data were initially recorded for 10 seconds at rest condition. After which, participants were told that they needed to play a VR game for 10 minutes and they would receive a reminder from the researcher when playtime was up. After the remainder, the participants could decide whether to stop playing or to continue playing for an extra 5 minutes. Each participant was informed that their EEG signals and eye movement would be recorded during playtime and they could ask the researcher any questions about the game or controls. After the VR gaming session, the LooxidVR cap was removed, participants were then asked to complete another attention test and filled out the post-experiment questionnaire including emotion reactions and immersion experiences. Finally, they were debriefed with the aim of the experiment before signing the subject receipt form and receiving the £10 Amazon Voucher or 0.5 SONA credit.

4 RESULTS

4.1 EEG

For EEG analysis, data from three participants were excluded due to poor EEG quality. For the remaining sample, the overall mean absolute beta wave power for HVGA group was $M=2.63$, $[SD=.68]$ whereas the mean value for LVGA group was $M=5.86$ $[SD=1.72]$. An independent t-test was used to compare mean difference between HVGA and LVGA groups. The results indicated that there was a significant difference $[t(25)=2.79, p<.001]$ in absolute data power between the HVGA and LVGA group. As for absolute total power, the mean value for the HVGA group was $M= 5.47$ $[SD=3.12]$ whereas the mean value for the LVGA group was $M= 5.12$ $[SD=2.65]$. An independent test was used to compare mean differences between the HVGA and LVGA groups. The results indicated that there was no significant difference $[t(25)=-1.49, p=.67]$ in absolute total power between the HVGA and LVGA. Details were presented in table 1.

Table 1: Differences in Absolute Beta Power and Absolute Total Power Between the HVGA and LVGA Group.

	HVGA		LVGA		<i>p</i>
	Mean	SD	Mean	SD	
Absolute Beta Power	2.63	.68	5.86	1.72	< .001
Absolute Total Power	5.47	3.12	5.12	2.65	.67

4.2 Perceived Immersion

For the self-reported immersion measure, scores of each participant were calculated by adding ratings for the 17 statements in the questionnaire. Since each statement had a maximum rating of 5, the overall scores were calculated out of 85, with higher scores

indicating greater immersive experiences. Across the sample, the overall mean immersive score for HVGA group was $M = 58.73$ [$SD=11.82$] whereas the mean score for LVGA group was $M = 5.94$ [$SD=6.32$]. For analysis, an independent t-test was used to compare immersion scores between HVGA and LVGA groups. The results indicated that there was no significant difference [$t(25)=.85, p=.403$] in immersion scores between the HVGA and LVGA group. Details were presented in table 2.

Table 2: Differences in Perceived Immersion Between the HVGA and LVGA Group.

	HVGA		LVGA		p
	Mean	SD	Mean	SD	
Immersion Scores	58.73	11.82	5.94	6.32	.41

4.3 Emotional Response

For the emotional response survey, scores of each participant were calculated by adding ratings for the 20 moods (10 positive moods and 10 negative moods) in the questionnaire. Since each mood had a maximum rating of 5, the overall mood scores were calculated out of 100, while the positive mood scores and negative scores were 50 respectively. Higher scores indicating greater emotional responses. Details were presented in table 3.

Overall Emotional Responses Across the sample, the mean overall emotional scores for the HVGA group was $M = 32.13$ [$SD=28.53$] whereas the mean scores for LVGA was $M = 28.53$ [$SD= 11.93$]. For analysis, independent t-test was used to compare overall emotional responses between HVGA and LVGA groups. The results indicated that there was no significant difference [$t(25)=.75, p=.46$] in overall emotional responses between HVGA and LVGA group.

Positive Emotional Responses Across the sample, the mean positive emotional scores for the HVGA group was $M = 25.07$ [$SD=8.56$] whereas the mean scores for LVGA was $M = 22.71$ [$SD= 8.30$]. For analysis, independent t-test was used to compare positive emotional responses between HVGA and LVGA groups. The results indicated that there was no significant difference [$t(25)=.79, p=.44$] in positive emotional responses between HVGA and LVGA group.

Negative Emotional Responses Across the sample, the mean negative emotional scores for the HVGA group was $M = 7.07$ [$SD=7.71$] whereas the mean scores for LVGA was $M = 5.82$ [$SD= 5.95$]. For analysis, independent t-test was used to compare negative emotional responses between HVGA and LVGA groups. The results indicated that there was no significant difference [$t(25)=.51, p=.61$] in negative emotional responses between the HVGA and LVGA group.

Positive and Negative Mood difference Paired t-tests were used to compare positive mood scores and negative mood scores within the HVGA and LVGA groups. The results indicated that there was a significant difference [$t(12)=11.40, p < .001$] between positive mood scores and negative scores in HVGA [$M=18, SD= 6.12$]. Consistently, there was a significant difference [$t(12)=8.55, p < .001$] between positive mood scores and negative scores in LVGA [$M=16.88, SD= 8.14$].

Table 3: Differences in Emotional Response Scores between HVGA and LVGA Group.

	HVGA		LVGA		p
	Mean	SD	Mean	SD	
Overall Emotional Response Scores	32.13	28.53	29.54	11.93	.46
Positive Emotional Response Scores	25.07	8.56	22.71	8.3	.44
Negative Emotional Response Scores	7.07	7.71	5.82	5.95	.61

4.4 Flanker Task Performance

In analysis of variance for Reaction Time (RT) and Accuracy in the Flanker task, Group (HVGA vs. LVGA) was a between-subject variable. Whereas phase (pre-VR gaming vs. post-VR gaming), was a within-subject variable. For each dependent variable, a 2 (group: HVGA vs. LVGA) $\times$ 2 (phase: pre-VR gaming vs. post-VR gaming) repeated measures ANOVA was conducted. Details were presented in table 4 and 5.

Response Time

Congruent trials: A 2 (group) $\times$ 2 (phase) repeated measures ANOVA revealed a statistically significant main effect of Group [$F(1, 24) = 5.20, p = .03, \eta_p^2 = .015$] indicating that the HVGA [$M = 440.48, SD = 53.55$] reacted faster on congruent trials as compared to the LVGA group [$M = 501.67, SD = 93.19$]. The main effect of Phase [$F(1, 24) = 7.77, p < .05, \eta_p^2 = .21$] was also significant, indicating that both groups reacted faster in the post-VR gaming test [HVGA : $M_{post} = 417.08, SD = 44.66$; LVGA : $M_{post} = 494.26, SD = 79.29$] as compared to pre-VR gaming test. [HVGA : $M_{pre} = 463.9, SD = 62.43$; LVGA : $M_{pre} = 509.08, SD = 107.09$]. There was no significant interaction of Group with Phase, [$F(1, 24) = 2.10, p = .16, \eta_p^2 = .07$] indicating that VR gaming did not have different effects for each group over time.

Incongruent trials: A 2 (group) $\times$ 2 (phase) repeated measures ANOVA revealed a statistically significant main effect of Group [$F(1, 24) = 9.51, p < .05, \eta_p^2 = .24$] indicating that the HVGA [$M = 490.66, SD = 25.16$] reacted faster on incongruent trials as compared to the LVGA group [$M = 591.37, SD = 20.81$]. There was no significant main effect of Phase [$F(1, 24) = 1.29, p = .27, \eta_p^2 = .04$], indicating that both groups reacted similar in the post-VR gaming test [HVGA : $M_{post} = 476.46, SD = 23.72$; LVGA : $M_{post} = 580.79, SD = 19.62$] as compared to pre-VR gaming test. [HVGA : $M_{pre} = 504.87, SD = 30.08$; LVGA : $M_{pre} = 591.96, SD = 24.88$]. There was no significant interaction of Group with Phase, [$F(1, 24) = 1.09, p = .30, \eta_p^2 = .04$] indicating that VR gaming did not have different effects for each group over time.

Accuracy Congruent trials: A 2 (group) $\times$ 2 (phase) repeated measures ANOVA revealed no significant main effect of group [$F(1, 24) = 1.09, p = .33, \eta_p^2 = .034$] or phase [$F(1, 24) = 0.082, p = 0.83, \eta_p^2 = .013$] on accuracy, or a significant group $\times$ phase interaction. [$F(1, 24) = 2.45, p = 0.18, \eta_p^2 = .073$], suggesting accurate responses for congruent trials did not change significantly across phases in both groups. These results demonstrated that VR-gaming session did not improve participants' accuracy in either the HVGA [$M_{pre} = 99.61, SD = 1.81$; $M_{post} = 99.80, SD = 4.32$] or the LVGA group [$M_{pre} = 98.61, SD = 9.72$; $M_{post} = 99.13, SD = 3.1$].

Table 4: Differences in Average Reaction Times and Accuracy for Congruent Condition between HVGA and LVGA Group.

Congruent Condition	HVGA		LVGA		<i>p</i>
	Mean	SD	Mean	SD	
Average Response Times (ms)	440.48	53.55	501.67	93.19	< .05
Average Accuracy (%)	99.71	3.07	98.87	6.41	.33

Table 5: Differences in Average Reaction Times and Accuracy for Incongruent Condition between HVGA and LVGA Group.

Incongruent Condition	HVGA		LVGA		<i>p</i>
	Mean	SD	Mean	SD	
Average Response Times (ms)	490.66	25.61	591.37	20.81	< .05
Average Accuracy (%)	98.21	2.87	96.46	7.93	.25

Incongruent trials: A 2 (group) $\times$ 2 (phase) repeated measures ANOVA revealed no significant main effect of group [$F(1, 24) = 1.37, p = 0.25, \eta_p^2 = .044$] or phase [$F(1, 24) = 0.079, p = 0.78, \eta_p^2 = .003$] on accuracy, or a significant group $\times$ phase interaction. [$F(1, 24) = 2.38, p = 0.14, \eta_p^2 = .073$], suggesting accurate responses for incongruent trials did not change significantly across phases in both groups. These results demonstrated that VR-gaming session did not improve participants' accuracy in either the HVGA [$M_{pre} = 98.61, SD = 1.51; M_{post} = 97.80, SD = 4.22$] or the LVGA group [$M_{pre} = 95.61, SD = 10.89; M_{post} = 97.13, SD = 5.1$].

4.5 Extra Playtime Findings

Differences were seen in the number of participants between groups requesting the 5-minute extra playtime. In particular, 9 out of the 13 participants in HVGA were noted to demand extra playtime after the required 10 minute play session. Additionally, 2 of the participants expressed willingness to continue playing when extra play time had been used up. In contrast, only 2 out of the 13 participants in LVGA requested the extra playtime.

5 DISCUSSION

5.1 Results Justification

The formulated hypothesis were as follows:

1. Differences in absolute beta power between HVGA and LVGA, with lower absolute beta power in HVGA group.
2. Differences in emotional responses between HVGA and LVGA, with greater emotional response in HVGA group.
3. Differences in perceived immersion between HVGA and LVGA, with higher perceived immersion in HVGA group.
4. Better selective visual attention performances in pre- and post-gaming attention test, with HVGA group reacted faster than LVGA group in both conditions.
5. Difference in number of participants requesting for extra playtime, with more requests from HVGA group

Based on result analysis, the first hypothesis was met as a significant difference was found in the absolute beta power between HVGA and LVGA with the HVGA group obtaining lower absolute

beta power in comparison to LVGA group. Previous research [7,59] suggested that lower absolute beta power might be a possible neurological biomarker of higher vulnerability for game addiction. The findings from this present study offered further evidence supporting the potential use of absolute beta power as a biomarker for the extent of game addiction. Moreover, no significant difference was found of total wave power between two groups suggesting that there might be another difference in other brain waves such as alpha, delta or theta waves. Indeed, difference in delta waves were also found to be lower in individuals with IGD when compared to the healthy control group in a former study [34]. As the present study only focused on beta waves, it remains unknown whether there would be any other difference in brain wave patterns during gameplay between the HVGA and LVGA group. Future research could be carried out to construct a more comprehensive model to provide guidance on game addiction diagnosis.

The second hypothesis was rejected as the results indicated that no significant difference was observed in the level of emotional response between HVGA and LVGA group. However, it might be worth noting that the HVGA group obtained higher mean values across all emotion categories, (overall, positive, and negative) yet the difference was not statistically significant. The findings of the current study opposed the previous research of emotion arousal, a possible explanation for this might be that the prior study had a participant pool that consisted of male participants only while the current sample comprised more female participants than male participants. This might affect the outcome as the literature to date suggested that male game players appear to be at greater risk of becoming addicted to games than female game players [65]. Nevertheless, the cause and severity of this demographic characteristics still remain unclear and hence the differing results might be a consequence of sampling bias. Additionally, as adolescents and young males have long been seen as the primary consumer in the current game market, and are thought to play video games more frequently than any other socio- demographic groups, gameplay design may also favour the males in the group resulting in gender bias altering the results [9]. Another possible explanation might be that game players in the previous research were excessive gamers rather than addictive gamers. As discussed above in the literature review, they are two distinctive groups and thus may react differently in terms of emotional response. As a result, more studies need to be done with a male- only sample to exclude gender difference. Moreover, as the emotional survey used in the current study was answered in self-reported manner rather than using EEG measurement, participants may recall incomplete or inaccurate emotional arousal when filling the survey after 10 minutes of gaming, as it is challenging for them to remember all the emotions they experienced during the game. Additionally, their self-reported answers might be affected by social desirability bias, which means they were hesitating to give their honest feedback when such answers are considered socially undesirable [45]. One significant finding from emotional survey was that both the HVGA and LVGA groups experienced significantly more positive emotions than negative emotions during gameplay, which further supported the idea that gaming could be used as an effective tool to ensure one's emotional wellbeing if used in appropriate channels.[16] Therefore, it might be valuable to raise awareness

of healthy gaming so that more people could enjoy quality leisure time without worrying about game addiction.

The third hypothesis was also rejected since there was no significant difference found in immersion scores between HVGA and LVGA. As there was no previous literature that had investigated the differences between the groups in perceived immersion, no comparison with prior studies could be discussed. Again, the HVGA group obtained higher average scores but not enough for it to be statistically significant. However, one possible explanation for this might be that, due to the nature of VR gaming, it was designed to offer a highly immersive and interactive gaming experience for the general public. As such the characteristics of VR games may differ from multiple player online games in terms of their core gameplay and reward mechanism. Indeed, given the fact that VR gaming has only been commercialised in recent years, no clinical case of VR game addiction has been reported so far and thus this area that seems to be under researched. However, one noteworthy research was carried out by Park et al in which they used Virtual Reality Therapy (VRT) as a treatment for Online Game Addiction (OGA). Their results signified that VRT tends to be effective in reducing the severity of OGA and showing similar treatment outcome when compared with Cognitive Behaviour Treatment. Moreover, the overall balance of cortico- striatal-limbic circuit was improved after VRT. Their study provided a new insight into how VR technology could be combined with the current treatment of game addiction to make it more effective. If more research could be conducted to investigate in greater detail the design principles of VRT gamification, findings of that research would be of great assistance in optimising currently used treatments for game addiction. Additionally, more accurate results could be acquired if eye-tracking data could be obtained and used as an index for between-group comparison like the original study had done [8].

The fourth hypothesis was partially met as reaction times for flanker task were significantly faster for the HVGA group than the LVGA group for all congruent trials in both pre- and post- gaming phase. This finding is consistent with previous research and supported the claim that video gaming could improve attentional capacity, and such enhancement tends to be even greater in HVGA group [13,51]. However, it failed to replicate the same effect for incongruent trials, as there was only a significant group difference, no significant difference was not found in reaction time between the pre- and post- gaming phase for incongruent trials. This is because for incongruent trials, HVGA performed similarly to LVGA group with regard to reaction times. According to Collins and Freeman, a number of studies failed to replicate such results and they explained that specific task feature might be responsible for generating consistent significant group differences. For example, Irons et al [24] suggested that the group difference could be due to aspects of the task design and paradigms where minor changes could result in insignificance. As the flanker tests used in the current study were self- designed rather than a standardised test, it might explain why consistent patterns were not observed.

Moreover, no significant effect of group or phase was found when it came to accuracy which means the two groups performed somewhat similarly across various trial types and phase conditions. This might be caused by the fixed difficulty of both flanker tasks as the correct response rates for both groups across all trials were quite

high. This may indicate the experimenter's self-designed flanker tasks were too simple and not validated enough to be used for measuring and comparing performances. However, Sun et al [62] advised that it is hard to interpret whether HVGA and LVGA groups should vary to a greater degree or even not to be consistent with the original prediction. They explained that group differences were more a lack of cognitive advantage derived from gaming instead of a reduction in performance. Therefore, it might be difficult to examine in the absence of a demonstrable cognitive advantage.

Finally, the last hypothesis was met as more HVGA participants requested for extra play session than LVGA participants. This suggests that HVGA participants may find it more challenging to withdraw from their highly engaged gameplay in comparison to LVGA participants, even if it was just the first contact with an unfamiliar game and the extra time granted was only 5 minutes. This finding is supportive of the view that highly engaged gamers with maladaptive coping styles may be at higher risks to develop video game addiction [2]. However, as no measure of game engagement was adopted in the current study, this finding lacks solid proof and more follow-up research is needed. Due to the limitation of a demanding experiment schedule, we did not manage to investigate whether there would be a difference in average playing time between HVGA and LVGA as play time for every participant is fixed. Future studies may take this factor into consideration as findings from current study suggested HVGA group tend to play longer.

6 CONCLUSION

To conclude, with the aim to explore latent factors associated with higher risk of game addiction, the present study set to investigate differences between the HVGA and LVGA groups in terms of their brain activities, emotional response, perceived immersion, and visual selective attention ability. The results showed that lower absolute beta power might be a neurobiological marker for greater vulnerability to game addiction. However, no significant difference was discovered in relation to overall emotional responses and perceived immersion between two groups. As for selective visual attention, the HVGA group was found to outperform the LVGA group with faster reaction times and relatively higher accurate response rates. Findings from the current study provided a deeper understanding in terms of the definition and diagnosis criteria of game addiction. As a new platform has been established which enables forthcoming studies to focus on assessing which neurobiological factors might be accounted to game addiction vulnerability. Accordingly, it could be the foundation for refining currently-used game addiction treatment. With the aid of neuroimaging techniques and machine learning algorithm, validation of diagnostic tools could also be updated.